EPFL - Computer Science

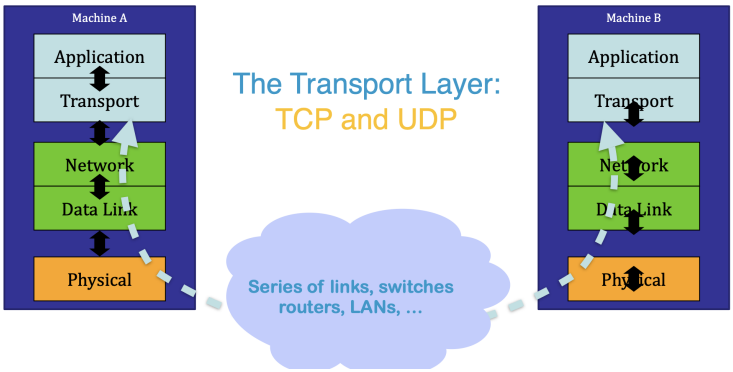
COM-402 Advanced Networks

Lecture 07 — November 27, 2025

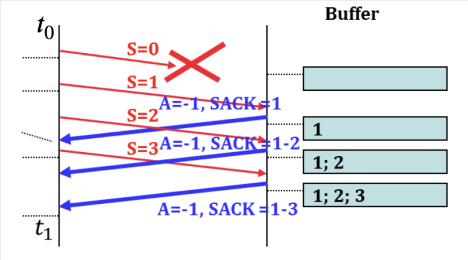
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Transporter Layer Continued



Window size = 4'000 bytes, one packet = 1'000 bytes. At time t_0, the usable part of the sliding window is 4000, and there is a lot of data in the socket to be sent. So, the sender sends 4 segments. At time t_1, sender...



A. Can send segment S=4
B. Cannot send segment S=4
C. It depends on whether data was consumed by application
D. I do not know

Answer: B.
Packet 4 it outside the window [0:3] and cannot be sent. It has to wait until the loss of packet 0 is repaired, which happens at time t_2

- Sender needs a *retransmission buffer* to hold unack'ed data; its size is equal to the window size.
- Segments are removed from the resequencing/receive buffer when they are finally put in order and application reads them.

TCP Sockets (simplified scenario)

Why both TCP and UDP?

Most applications nowadays use TCP, as this avoids re-inventing error recovery for each app. But some apps may not need TCP-like error recovery (i.e., packet retransmission):

- Voice applications / Sensor data streaming
- An application that sends just one message, like name resolution (DNS)
- Multicast (TCP does not support multicast IP addresses)

4. More TCP Bells and Whistles

How to deal with **corner cases**? Suppose that an application writes few bytes in socket; should TCP create a segment or wait for more bytes?

- Nagle's algorithm: send `min{bytes written in one RTT, MSS}`

How to handle silly window-increase advertisements by 1 byte? The idea is similar to Nagle's, we do something simple

Window field is 16 bits $\Rightarrow$ ma throughput even for RTT=1msec is $524 Mbps; can we do better$?

- negotiate a scaling factor of the window during connection setup (RFC1323)

How to detect packet loss?

Packet loss in TCP is detect via times and duplicates ACKs.

The main timers are called retransmission timeouts (RTO), it's set for every packet transmission and when a timer expires (= timeout), this is interpreted as severe loss in the channel. Thus, all timers are reset and all un-acknowledged data is marked as needing retransmission

A duplicate ACK = a TCP segment where the cumulative ACK value repeats a previously received ACK value. This is interpreted as not-so-severe loss in the channel.

Timers use an estimate of round trip time (RTT)

Why? RTO timer must be set to a value $\geq$ RTT, but not too much

What? RTT estimation gives an upper bound on the round trip time, used to set the RTO

How?